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Entropy-based Sentence-level Hallucination Score in Large Language Models

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***Abstract*—Large Language Models (LLMs) excel at generating human-like text but are prone to hallucinations—fabricated outputs that deviate from reality. Addressing this is critical for ensuring trustworthy AI, especially in fields like medicine, law, and education. This paper introduces an entropy-based metric for detecting sentence-level hallucinations by assessing token prediction uncertainty. High entropy correlates with hallucinatory outputs, and we validate this metric's effectiveness across different LLMs. Applications include hallucination flagging and improving user trust through uncertainty-driven feedback. This approach advances the reliability and transparency of LLMs, contributing to safer AI systems.**

***Index Terms*—Deep Learning, Natural Language Generation, Large Language Model, Hallucination Detection**

I. Introduction

Large Language Models (LLMs) have advanced natural language processing (NLP) by enabling fluent and contextually relevant text generation for tasks like translation, summarization, and question answering. Models such as OpenAI's GPT, Google's Gemini, and Meta's LLaMA demonstrate near-human performance [1]–[3], leveraging transformer architectures to process extensive datasets and generate human-like responses. However, LLMs often produce outputs that appear credible but are factually inaccurate or fabricated, posing risks in critical fields. Addressing these issues is essential for building reliable AI systems. Previous methods to address hallucinations include semantic consistency checks and uncertainty quantification (UQ). SAC^3, for instance, compares outputs from varied prompts to detect inconsistencies [4], while UQ-based methods like POPE and LURE flag unreliable responses by assessing prediction uncertainty [5], [6]. These techniques underscore the importance of combining semantic validation with uncertainty metrics to enhance LLM reliability.

We propose the Entropy-based Sentence-level Hallucination Score, which uses token-level entropy to identify sentences at risk of hallucination. Higher entropy signals greater uncertainty, marking unreliable outputs. This method offers precise, real-time detection, contributing to safer and more dependable AI systems.

II. Proposed Method

In [7], entropy is introduced as a measure of predictive uncertainty in conditional language generation, quantifying a language model's uncertainty about the probability distribution of possible tokens at each generation step. High entropy reflects greater uncertainty, often correlating with a higher likelihood of hallucinations, as the model lacks confidence in selecting specific tokens.

The entropy $H(y_i|c_i)$ for predicting a token y_i given context c_i is defined as:

$$H(y_i|c_i) = -\sum_{v \in V} p(y_i = v|c_i) \log p(y_i = v|c_i) \tag{1}$$

where V is the token vocabulary, and $p(y_i = v|c_i)$ is the probability of token v at position i given the context c_i.

For our work, responses generated by the LLM were split into sentences using SpaCy [8]. Sentence entropy, calculated as the average of token-level entropies within a sentence, is defined as:

$$H(Sen_j) = -\frac{1}{N_j} \sum_{k=i}^{i+N_j} p(y_k = v|c_k) \log p(y_k = v|c_k) \tag{2}$$

where N_j is the token length of the jth sentence (Sen_j). This metric aggregates uncertainty over all tokens within a sentence, offering a sentence-level measure of predictive uncertainty.

III. Experiment

A. Experiment Result

In our experiment, we evaluated large language models using three datasets—QAMPARI, FAKEQAMPARI, and TriviaFacts—comprising 297 samples, following the setup from [9]. Table I highlights that hallucinatory sentences consistently exhibit higher entropy than non-hallucinatory ones, with sentence-level entropy offering better differentiation. For example, in the FAKEQAMPARI dataset, Qwen2 [10] shows a token-level entropy gap of 0.5120 (0.9630 vs. 0.4510), while the sentence-level gap increases to 0.6161 (1.0613 vs. 0.4452). Similarly, in TriviaFacts, Gemma2 [11] shows a token-level gap of 0.4640, expanding to 0.4826 at the sentence level.

The larger gap at the sentence level demonstrates its superior ability to differentiate between hallucinations and accurate

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TABLE I
THE AVERAGE OF SENTENCE ENTROPY IN EACH EVALUATION DATASET

Dataset	Model	Non-hallucinate (Token)↓	Non-hallucinate (Sentence)↓	Hallucinate (Token)↑	Hallucinate (Sentence)↑
TriviaFacts	LLaMa3	0.2301	0.2203	0.5886	0.5944
	Qwen2	0.4513	0.4239	0.9602	1.0027
	Gemma2	0.1893	0.1919	0.6533	0.6745
QAMPARI	LLaMa3	0.2554	0.2243	0.4815	0.4988
	Qwen2	0.4316	0.4125	0.8286	0.8114
	Gemma2	0.1919	0.1964	0.5807	0.6020
FAKEQAMPARI	LLaMa3	0.1451	0.1416	0.5929	0.6312
	Qwen2	0.4510	0.4452	0.9630	1.0613
	Gemma2	0.1635	0.1454	0.8184	0.7789

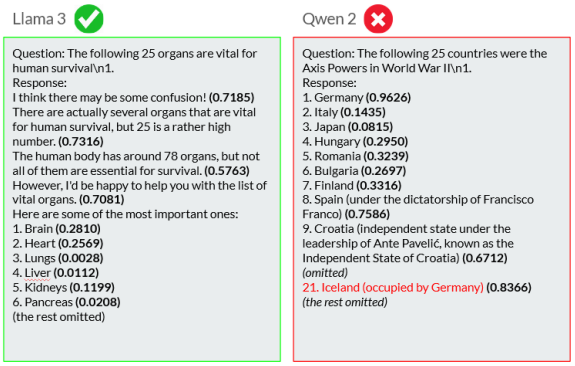


Fig. 1. Example of high entropy in a non-hallucinatory response (left) and low entropy in a hallucinatory response (right), illustrating challenges in distinguishing hallucinations using entropy metrics alone.

outputs. By aggregating token-level uncertainty across a full sentence, sentence-level entropy proves to be a more effective tool for hallucination detection, particularly in capturing subtle differences in model behavior.

B. Discussion

Figure 1 compares sentence-level and token-level entropy for hallucination detection on the TriviaFacts dataset.

In the left example, LLaMA3 provides a correct response. Sentence entropy (0.3789) correctly classifies it as non-hallucinatory, being closer to the non-hallucinatory threshold (0.2203), while token entropy (0.4535) incorrectly suggests hallucination. This demonstrates sentence entropy's better reliability by capturing contextual uncertainty.

In the right example, Qwen2 hallucinates by listing non-Axis countries, but both metrics fail to detect it. Sentence entropy (0.5951) and token entropy (0.6661) remain closer to non-hallucinatory thresholds.

These cases highlight sentence entropy's strengths over token entropy but also reveal the challenges in detecting nuanced errors.

IV. Conclusion

We introduced an Entropy-based Sentence-level Hallucination Score to assess the likelihood of hallucinations in LLM outputs. Experiments on three datasets show that sentence entropy effectively reflects model uncertainty, often aligning with hallucination instances. Higher entropy values are frequently associated with hallucinations, demonstrating its potential as a detection metric. However, not all hallucinations occur in high-entropy sentences, and some high-entropy sentences are accurate. This suggests that sentence entropy, while valuable, should be paired with additional methods to improve reliability in hallucination detection.

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